Supporting menu design with radial layouts

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ABSTRACT
One of the ways of decreasing selection times in displayed menus is through the use of radial layouts which shorten distances to the items and increase the item sizes. Previous work on radial menus does not demonstrate that radial layouts are beneficial for menu hierarchies. The paper focuses on this topic.

We compare two main characteristics of radial and linear layouts: a) the time it takes to find an item (i.e. visual search time), b) the time it takes to navigate to an item (i.e. pointing time). We use objective and subjective measures, two menu sizes, three menu levels, one linear and three radial layout variations.

We also present requirements and the Compact Radial Layout (CRL) design which supports menu hierarchies. We assess its performance for novice and expert users.

Categories and Subject Descriptors
H.5.2 [**User Interfaces**]: Interaction styles.

General Terms
Design, Performance, Measurement.

Keywords
Menus, Radial Menus, Radial Layout, Visual Search, Pointing.

1. INTRODUCTION
Menus are one of the primary GUI controls providing both information on what commands are available and a way to invoke them. As software applications continue to grow in terms of the number of features they offer, menus have grown in size, and have become both hierarchical and more difficult to use. Usually each new release of an application provides additional functionality. Consequently, menu designs have to accommodate a large number of functions. Providing easy and fast access to these is a real challenge. The issue is especially important for professionals who utilize menus extensively, invoking hundreds of commands during their normal work session (e.g. 3d modelers or video editors).

A set of cascading menus is one of the most widely used designs to access hierarchies of commands. For the user, the typical view of the path to access a particular command appears as a sequence of linear menus glued together. Each submenu must be navigated in turn as a user performs a selection. The cost of navigating multiple levels to select an item is high. In particular, moving between submenus is time consuming and error prone [2].

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One approach to decreasing the navigation requirements in menus is to use alternative layouts (i.e. different spatial placement of the menu items). Radial layout is one example that both shortens the distances between the menu items and increases their size. Despite the proven potential of a radial layout for single level menus, not much research has explored its efficiency for command hierarchies.

This paper contributes a design of hierarchical radial menu and a study investigating how radial layout influences two main factors governing menu selections – searching for menu items and pointing to menu items. We use objective and subjective measures. We also use collected data to assess novice and expert performance of tested designs.

The rest of the paper is structured as follows. First we present related work exploring the radial layout for displayed menu design. Then we present our hierarchical radial menu design and its goals. An experiment follows in which we investigate the demands posed by the radial layout on visual search and pointing. Next, we report the results, discuss the findings and close the paper with the conclusions.

2. RELATED WORK

2.1 Background
In the context of displayed menus, radial layouts were initially used to support small, single level menus. Support for hierarchies was first realized by a series of single level radial menus. Work explicitly utilizing radial layout for menu hierarchies came afterwards. These three categories are discussed in the following sections.

Novices unfamiliar with the menu content and command locations have to search for commands visually. In this case, the overall selection time depends on visual search and pointing times [7]. However, users quickly develop and rely on spatial knowledge for menu item locations (i.e. they become expert users) if the layout of the menu is stable [12, 19]. Consequently, visual search time decreases as users become experts and the overall selection time then depends mainly on decision and pointing times [7, 16].

2.2 Single level radial menu
Radial layout was recognized early as a technique for decreasing pointing time in menu design. Pie menu (PM) [5] involves placing the items along a circumference of a circle at equal distances from the center. Each item contributes 'a slice of pie'. To perform a selection the user needs to move the cursor only by a small amount towards an indented item. The design supports only single level menus and can only accommodate a small number of items effectively. In a controlled experiment, PM was compared to a linear menu in expert mode, using 8 item menus, yielding 15% faster selections. Consequently, radial layouts decrease pointing time for expert users, but it is not clear how

efficient non-expert users would be when scanning the ring to search for items.

2.3 Series of single level radial menus

The topic of supporting menu hierarchies with radial layouts has appeared in the context of marking menus [13, 14, 20]. Marking menus extend PMs with an additional selection mode for experts. In this additional mode, the user can issue a straight 'mark' in the direction of a desired item without the actual menu being displayed. Selections from hierarchies of commands result in a series of marks uniquely identifying a menu item. Only expert users that remember the exact spatial locations of frequently used commands can benefit from this mode. If the user does not remember the location of a desired item, or hesitates in his mark, a menu is displayed and selection can be made in the traditional way. Each level in the displayed menu is represented as a separate PM and must be navigated individually. There are two strategies for laying out a series of pies:

- **Overlapping Pie Menus (OPM)**. If the pies do overlap the distances between the levels are shorter but the access to the previous levels is hindered. As such, OPM are beneficial for expert users who know the exact or approximate item locations and appreciate the decreased navigation effort. However, OPM inconveniences novices and non-frequent users who, while searching for the right item, tend to traverse the menu hierarchy back and forth.
- **Non-overlapping Pie Menus (NPM)**. The situation is reversed if the pies are repositioned to avoid overlaps. The previous levels are visible and accessible but the distances between the levels as well as the required screen real-estate are increased.

As such, a mere sequence of single PMs does not accelerate access to all menu items and levels. Additionally, it is only evident that PM (either with or without overlaps) is faster than a linear menu layout for a small number of items (i.e. 8) [5]. Increasing the number of items will decrease each item's size and will require the pie radius to be increased, which also effectively increases the distance to each item. Smaller and more distant items take longer to select [18]. Therefore it is questionable how effective PM could be for larger menus. Eight items on each level is too restrictive in many cases as menu hierarchies often contain an increased number of items on the second or third level. The research on marking menus focuses on mark efficiency. The actual performance of selecting from a displayed sequence of pies was not studied nor compared to the typical linear menu. Therefore the merit of using pie series as a primary, displayed menu mode has not been established.

2.4 Hierarchical radial menu

A design explicitly leveraging radial layout for displayed menu hierarchies was presented in [3]. The primary design – Wave Menu (WM) - expands inward showing the items on subsequent levels in the innermost ring. The goal is to decrease navigation and visual effort by keeping the focus always in the center of the menu. The main drawback of WM is the limited space on the first ring (similar to PM) and the lack of positional constancy for the already selected items as they move to the outer rings. An alternative design (WM2), where the subsequent levels appear in the outer rings, was also shortly mentioned in the paper. In both designs, the menu items are spread around the whole ring and are represented by equal-sized segments of the ring. The experiment presented in [3] compared four radial designs: WM, WM2, OPM and NPM designs. Only novice behavior was simulated. The menus had two levels with 4, 6, or 8 items each, thus containing 16 to 64 items in total. The results proved that OPM hinders access to previous levels and is slower for novices who employ extensive search (e.g. exhaustive search). However, the paper did not compare presented radial designs to linear organizations (e.g. deployed in cascading pop-up menus). Moreover, expert behavior was not simulated and therefore it was not evaluated if a radial layout enabled faster access to command hierarchies. Last but not least, the only design capable of accommodating an increased number of items on the second level (WM2) yielded the worst performance (no different from OPM), questioning the merit of using radial layout for larger menu hierarchies.

Hotbox is a hybrid pop-up menu design [15]. The first level items are organized in a way that resembles radial organization. The subsequent levels are accessed through cascading menus. One of its primary goals is to accommodate a large number of commands. The paper focused on design and did not include any experiment.

In summary, previous work [3, 5, 15] indicates that radial layouts may be beneficial for displayed menu design. However, it does not demonstrate the merit of using radial layout for menu hierarchies when both novice and expert behaviors are considered. This paper contributes a hierarchical radial menu design and a study investigating how radial layout influences visual search and pointing. The results are used to assess novice and expert performance of the proposed design.

3. DESIGN

3.1 Design principles

As the number of commands increases, operating an application becomes a more complex task. Traversing a three-level cascading menu in search of a desired command demands precise mouse movements and is time consuming even for experts. To accelerate user performance, systems typically include an additional menu access technique, such as shortcuts or marks, allowing the users to issue commands faster. However, these additional techniques rely on recall rather than recognition. The result is that users forget the command shortcuts and have to switch back to the original displayed menu. Moreover, an increased number of shortcuts are hard to remember and to articulate (e.g. Alt-Ctrl-x or physically similar marks).

Access to menus can also be accelerated through toolbars. Although visual, toolbars hinder the learning and finding of commands since they break menu organization strategy [15]. They also increase screen consumption.

In general, the necessity of having to learn two separate techniques to operate an application efficiently adds to the overall complexity.

We aim at creating a displayed menu design that is convenient for novices and fast for experts. Consequently, our goal is to create a single technique unifying novice and expert modes. Our design requirements are the following:

R1 *Quantity and space consumption*. Contemporary applications often contain hundreds or even thousands of commands, typically organized in cascading menus of depth 2, 3, or even more. A requirement is to accommodate a high number of

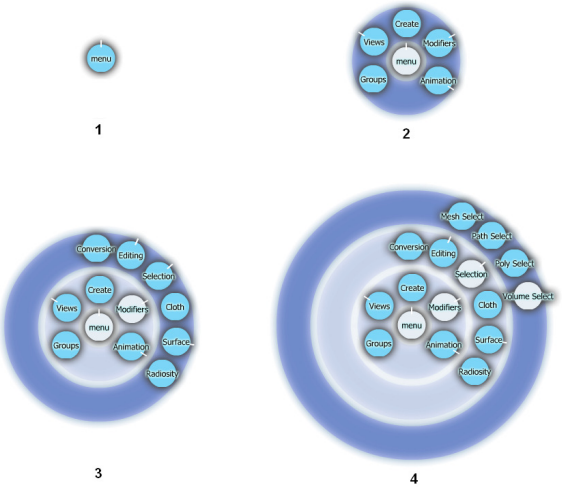


Figure 1 CRL design (4) and an example selection sequence.

commands by supporting at least three levels with low screen space consumption.

R2 *Fast access for experts*. To overcome the problem of slow access to menu items, a requirement is to support expert users by decreasing navigation time, not only on the first but all the levels and between the levels.

R3 *Facilitating exploration for novices*. To support novices, a requirement is to show the full context of the current selection sequence, and ease access to the content of the previous menu levels. The design also needs to enable fast visual scanning of the items on each menu level.

R4 *Smooth novice-to-expert transition*. Last but not least, the design needs to facilitate learning in order to accelerate novice to expert transition.

In the next section, we present a design based on the above requirements.

3.2 Compact Radial Layout Design

Figure 1 depicts the Compact Radial Layout (CRL) menu design and an example of a selection sequence. The levels of the menu hierarchy are represented as concentric rings. The first ring corresponds to the first level of the menu hierarchy. Subsequent rings, surrounding the previous ones, correspond to next levels of the menu hierarchy and represent a submenu of a selected parent item from the previous ring. Each subsequent ring has a larger circumference than the previous ring and therefore can accommodate more menu items (requirement R1). All parent items are marked with the line coming out of the item. There is a number of ways the items can be distributed within a ring (see Figure 2b – CRL, FAN, SPARSE). The SPARSE design distributes items evenly within a ring similarly to PM and WM/WM2 menus. The FAN design guides visual search by placing the first item from a submenu on the next outer ring in the closest position to the parent item. All further menu items are placed clockwise around the outer ring. In contrast to other radial designs, the CRL design packs items tightly around their parent, keeping the distances between items and levels as short as possible, thus allowing for fast selections and navigation (req. R2). Consequently, the length of the visual scan path on each ring is also shortened and space consumption is low (req. R4 and R1). Items can be easily counted and separated visually as they are represented by individual, disjoined circles rather than ring segments as in the case of PM and WM/WM2 radial designs. Note that this way, the rings in the CRL design do not have to be explicitly drawn, further reducing space consumption (req. R1).

The interaction with a CRL menu works as follows: the user clicks on the central item representing the CRL menu. The first ring surrounding the central item appears. The user then points to a desired item and selects it with a mouse click. If the item has no submenu, the command is invoked and all the rings disappear. If the item does have a submenu, the next outermost ring is displayed. Similarly, any subsequent rings will appear if the user selects a parent item placed on the outermost ring. The user can select from any visible items (req. R3). If a parent item on an inner ring is selected, all outer rings disappear and a new outer ring appears. Since all the submenus leading to the last submenu are visible the context is preserved (req. R3). Regardless of the user interaction, the locations of the menu items and rings never change. This assures positional constancy of the layout facilitating the novice to expert transition [7, 12, 19] (req. R4).

3.3 Design issues and alternatives

Several design issues arose when we tested and deployed CRL in real-world settings.

Limited space on the first ring. Although the CRL design can accommodate an increasing number of items on subsequent levels, the spatial capacity of the first level may be too small. Depending on the application, a combination of the following remedies can be used: a) smaller item (i.e. circle) sizes; b) changing the structure of the menu, such as adding an intermediate first level with fewer items; c) moving away the first ring from the central item by a desired amount, increasing its spatial capacity.

Menu location. As a CRL menu can expand in all directions, it is undesirable to place it close to the screen borders. Doing so may partially obscure the menu or require repositioning. For these reasons, we recommend using a CRL menu similarly to context menus typically invoked within the working space in the center of the screen [15]. This increases selection performance by not requiring time consuming round-trips to toolbars or tool pallets, which additionally pull the users away from their work. Alternatively, if the menu structure allows, the CRL menu can sit close to the corner of the screen and expand as a slice of pizza toward the center when navigated.

Long labels. Close proximity of the items to each other creates a possibility of overlapping labels. This situation can be handled in a number of ways: a) the circles can be enlarged to increase their spatial capacity; b) the distances between the items can be increased; c) long labels can be split into two or more lines. Note also, that the circular shapes of CRL's menu items are better suited for using icons than narrow cascading menu boxes.

Adaptation. The size of the circles can be emphasized or underemphasized based on an item's frequency of use, number of children, context-determined relevancy or other properties. This will not break positional constancy of the menu items within certain limits, and thus is beneficial for adaptive menu techniques.

4. EXPERIMENT

Previous work focused mainly on "design versus design" studies in novice or expert scenarios. The objective of this experiment is

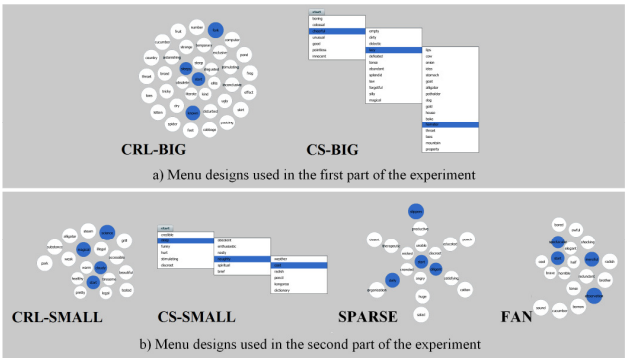


Figure 2 Menu designs used in the experiment.

to investigate how radial layout compares to linear layout in terms of visual search and pointing performance. The experiment includes two menu sizes, three levels, one linear and three radial menu variations. We are interested in objective and subjective measures. Finally, the collected data will serve to assess novice and expert performance of tested designs.

In order to measure the time required for visual search, we need to distinguish between visual search and pointing times.

4.1 Separating Visual Search from Pointing

In typical menu selection experiments, visual search and mouse movements times are measured as a single value. For this reason it is not clear how much of the total time should be attributed to either visual search or pointing. To separate both, we adopted a point-completion deadline method proposed in [10]: "*Once the mouse starts to move, the participant is given a limited amount of time to click on the target; if the participant does not click the mouse button before this point-completion deadline is reached, then the trial is interrupted and recorded as an error. Thus, participants are discouraged from moving the mouse until they have found the target, and the start of the mouse movement consistently marks the end of visual search.*" For our experiment, the point-completion deadline times were computed using Welford's version of Fitts' law with constant *a* and *b* set to 300 as suggested in [10]. The item widths required by Fitts' law were measured along the line of approach as suggested in [18].

4.2 Menus

The CRL menu, its two variations and one linear menu were used in the experiment. The cascading pop-up menu (CS) was chosen to represent typical linear context menus. The implemented menu is equivalent to those found in contemporary applications, such as MS Word, with two exceptions: 1) an explicit click on each level is necessary to open the submenu (i.e. submenus do not open after dwelling on a parent item); 2) diagonal movements from a parent item toward its submenu are possible without a risk of the submenu closing. These two exceptions eliminated an additional factor (click or dwell) from the experiment, kept the interaction technique for linear and radial menus the same, and allowed us to focus on the layout differences. Furthermore, while the search times will be the same, the pointing times for our CS menu will be lower than for other CS variants using dwelling and/or restricting diagonal movements. This is because mouse clicks are faster than typical dwelling times (~200ms [6] versus 333ms) and diagonal movements are faster than the combination of horizontal and vertical movements [2]. Therefore, we can safely assume that our CS design yields faster selections than other typical CS designs which do not have the mentioned exceptions. Consequently, we can use measured times as a lower bound for typical CS designs.

To determine the size of the items in the CS menu, we analyzed several menus of widely used applications and found that widths varied significantly. We chose the size of 190x21 pixels (area=3990 pixels) for the CS menu because this size could accommodate three word labels. To determine the size of the items for the radial menus, the volume of our circular items was not chosen to match the volume of our representative cascading menu item. Rather, we experimentally determined the size for the circles such that they provided enough space to also accommodate three word labels (eventually broken to two or three lines) while avoiding overlaps. The diameters for the circular items were 44 (a=1134p), 50 (a=1521p), and 54 (a=1810p) pixels for the first, second and third level respectively.

4.3 Task

The menu content was constructed in the following way: the first and second levels contained adjectives from a dictionary of 600 commonly used adjectives; the third level contained nouns from a dictionary of 400 commonly used nouns.

The task was to select a three word phrase of the form *adjective-adjective-noun*, such as *"cloudy magical science"*, from a three-level menu. The target phrases were constructed this way to make them easy to remember. Arbitrary three word phrases may be hard to memorize and recall during the course of the trial, and may negatively affect the measured times. Additionally, our approach allowed us to easily create content with an arbitrary menu size and desired degree of randomness. To prevent any learning effects and force participants to visually search for the items, a new random target phrase and menu content was generated for each trial and for each participant independently. Consequently, our task simulated novice behavior.

First, the target phrase was shown on a blank page. After reading and memorizing it, the participant proceeded with a mouse click. At this point the phrase disappeared and the menu was displayed. The target phrase disappeared in order to prevent rescans which would bias the visual search results [1]. The participant performed a task consisting of three selections, one word on each level.

The experimental setup imposed a point-completion deadline. In contrast to the study presented in [10], our experiment exploits this technique in a task comprising a series of selections. On each level, once the participant moved a mouse pointer by more than 5 pixels, he or she had a limited time for completing the next selection. If the participant did not select the next menu item before the deadline, the trial was stopped, a buzzer sound played, an alert message appeared reminding the participant to find an item before moving the mouse, and the trial was recorded as an error. After selecting an appropriate item, a new level appeared and the participant had to again find the next item before starting to move the mouse.

4.4 Design

4.4.1 Part 1 – BIG MENUS

The first part of the experiment included one radial and one linear menu, both containing 7, 12, and 17 items on the first, second and third level respectively (1428 items in total). This configuration fully populated all three rings in the radial menu creating a

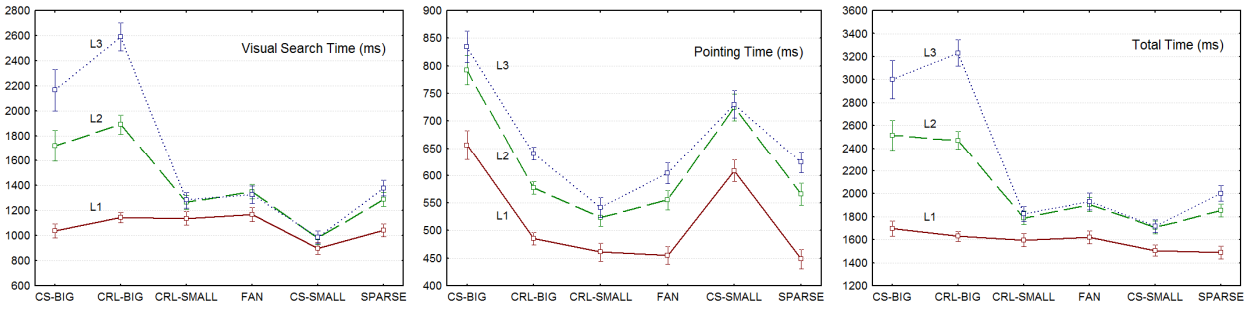


Figure 3 Visual search, pointing, and total time (+/- standard error) as a function of menu design and level (L1, L2, and L3).

situation where participants had to possibly search the entire circumference of the ring to find a menu item (see Figure 2a).

The sequences for the target phrases were randomly generated for each participant, assuring that the physical selection path lengths were evenly distributed within the range of all possible lengths. This way we did not promote easy or complex tasks on per participant basis. Each participant completed two blocks of 40 trials, one block for each menu. Half of the participants started with the radial menu and the other half with the linear menu. Tasks were presented to each participant in a random order.

4.4.2 Part 2 – SMALL MENUS

In the second part of the experiment, we used a smaller, more realistic menu configuration containing six items on all three levels (216 items in total). We used one linear and three previously introduced radial menu variations (see Figure 2b, and section 3.2) to test if the menu size and different techniques of laying out the items around the rings affect the performance.

Each participant completed four blocks of 18 trials, one block for each menu. For each menu type, we added two extra warm-up tasks which were excluded from the analysis. We used a Latin square to control the menu type order. Tasks were presented to each participant in a random order.

4.5 Measures

The experiment measured the following quantitative dependent variables: a) visual search time – measured between a mouse click and the following mouse move; b) pointing time – measured between beginning of a mouse move and the following mouse click; c) number of errors resulting from mouse clicks missing the targets. For each task, the visual search and pointing times were recorded for all three selections, one for each level.

For the qualitative analysis we administered two questionnaires measuring subjective attitude of the participants, and a free form interview. The first questionnaire was administered after the first part of the experiment. Participants were reminded that the task consisted of: a) finding an item and then b) pointing and selecting it. Then, they had to choose a menu for which **finding** an item was **easier** (first self-reported variable) (linear, radial, no difference) and a menu for which finding an item was **faster** (second variable). Similarly, participants had to choose a menu for which **selecting** was **easier** (third variable) and **faster** (fourth variable). A textual explanation had to be provided for all the answers.

The second questionnaire was administered after the second part of the experiment. The participants had to rank four menus on a 1-5 Likert scale according to two questions: *"How easy was it to **find** items using the menu X?"* and *"How easy was it to **select** items using the menu X?"*. Further, the participants had to order four menus according to their preference for the tasks performed. The questionnaires were anonymous.

4.6 Participants and apparatus

A total of 18 participants (5 female and 13 male) were recruited for the experiment. None of them had used the CRL menu before. All were between the ages of 22-30 and used a computer and a mouse on daily basis. All participants received sweets.

The experiment was conducted on an IBM T61p laptop running MS Windows XP. The connected monitor was a 20" IBM ThinkVision L200p LCD and used its native resolution of 1600x1200. A mouse was used to perform the tasks. The whole environment resembled typical office conditions.

4.7 Procedure

The average session lasted approximately 40 minutes per participant. The procedure was as follows. 1) The participants were briefly introduced to the experiment and explained that they should select an item only after finding it visually. 2) In a practice session participants became comfortable with the task. 3) The first part of the experiment started followed by the first questionnaire. After a short break they proceeded to the second part of the experiment followed by the second questionnaire. 4) The participants were interviewed. They were encouraged to freely express their comments, observations and suggestions.

4.8 Results

The results are graphically presented in Figure 3. For clarity, standard deviations (SD) were removed from the graphs. We present the results from both parts of the experiment together. SMALL and BIG are abbreviations referring to the first and second part of the experiment where the menus used were small and big respectively (see Figure 2). In both parts, menu ordering did not have any effect on the results.

To determine if there were differences in performance between the menus in the first part of the experiment, 2x3 (2 menus x 3 levels) Two-Way Repeated Measures (RM) ANOVA and post-hoc pair-wise comparisons using Bonferroni adjustment were calculated separately for visual search, pointing and total (visual search + pointing) time.

Although RM ANOVA is a powerful statistical tool, a set of assumptions have to be met to justify its use. For each dependent measure, and each part separately, we calculated a ratio between the highest and lowest SDs of means from all conditional groups. The maximal ratios varied between 1.6 and 3.1. RM ANOVA is not robust against the violation of assumption of equal variances. In such a situation, it is sensible to transform measured times in order to equalize the variances and obtain times that better

approximate normal distribution within conditional groups. We considered log and 1/x transformations. The log one was chosen because it equalized SDs much better.

The assumption of data sphericity was checked with Mauchly's test. When necessary we used the Greenhouse-Geisser correction.

For the data recorded in the second part of the experiment, we used 4x3 (4 menus, 3 levels) Two-Way RM ANOVA. The data was log transformed for the same reasons as in the first part of the experiment. Again, all post-hoc pair-wise comparisons used the Bonferroni adjustment, and the sphericity assumption was tested.

4.8.1 Visual Search Time

SMALL menus. There was a significant main effect for *menu* ($F_{3,51}$=78.4, p<.001, η^2=.82), *level* ($F_{2,34}$=52.8, p<.001, η^2=.76), and *menu x level* interaction (F6,102=4, p=.001, η^2=.19). Pair-wise comparisons show that CS (957ms SD=273) was consistently faster on each level (by 292ms on average) from all radial alternatives (avg=1249ms SD=338) which did not differ from each other.

Interestingly, however, while the SPARSE menu is the fastest radial alternative on the first level it becomes slower than the CRL menu on the second level and on the third level it becomes the slowest one. Although this pattern is clearly visible in the data and explains the interaction effect, it is not emphasized by the results of pair-wise comparisons.

Despite the same number of items on all three levels, the visual search times on the second and third levels, although strikingly similar, differ from the first level result (by 130ms). This pattern is visible for the CRL, FAN and CS menus. The reason could be the increased effort of having to move the eyes between the levels.

BIG menus. There was a significant main effect for *menu* ($F_{1,17}$=20.9, p<.001, η^2=.55) and *level* ($F_{2,34}$=274.6, p<.001, η^2=0.94). Times averaged across the levels were 1643ms (SD=572) for CS and 1876ms (SD=618) for CRL. Pair-wise comparisons show that the CS menu was faster than the CRL menu on the second level (by 172ms) and the third level (by 425ms) but not the first level.

4.8.2 Pointing

SMALL menus. The data was non-spherical for *menu x level* (χ^2(20)=48.5, p<.001, G-G ε=0.5). There was a significant main effect for *menu* ($F_{3,51}$=90.3, p<.001, η^2=.84), level ($F_{2,34}$=166.4, p<.001, η^2=.91), and *menu x level* interaction ($F_{3,51.3}$=6.7, p<.001, η^2=.28). Pair-wise comparisons show that all radial designs (avg=531ms SD=101) were faster from CS (688ms SD=129) on all levels and differed from each other only on the third level, where the CRL menu was found to be the fastest. The data shows a clear pattern where the CRL menu gets faster on each level

BIG menus. There was a significant main effect for *menu* ($F_{1,17}$=248, p<.001, η^2=.94) and *level* ($F_{2,34}$=144, p<.001, η^2=0.89). The comparisons show that the CRL menu (avg=568ms SD=97) compared to the CS menu (avg=761ms SD=152) was faster on all three levels (by 193ms on average).

4.8.3 Total time (Visual search + pointing)

SMALL menus. There was a significant main effect for *menu* ($F_{3,51}$=10.6, p<.001, η^2=.38), *level* ($F_{2,34}$=139.5, p<.001, η^2=.89), and *menu x level* interaction ($F_{6,102}$=7.8, p<.001, η^2=.31). Pair-wise comparisons show that on the first level the SPARSE menu is faster than the FAN menu (by 132ms). On the second and third levels, times for the SPARSE menu increased the most as a consequence of its increasing visual search time component. On the second level there is no significant difference between the radial menus, but only the CRL menu is not slower than the CS menu (the CS menu being faster than the FAN and SPARSE menus by 173ms). On the third level again, only the CRL menu is not slower than the CS menu (the CS menu being faster than the FAN and SPARSE by 250ms). Additionally, on the third level the CRL menu is faster than the SPARSE menu (by 180ms).

BIG menus. There was a significant main effect of *level* ($F_{2,34}$=341.8, p<.001, η^2=.95) and *menu x level* interaction ($F_{2,34}$=5, p=.01, η^2=.23). There was no main effect of *menu*. The interaction is present because the CRL menu has lower times than the CS menu on the first and second levels but higher on the third level. The pair-wise comparisons, however, show that menus do not differ significantly on the first, second and third level.

4.8.4 Questionnaires

SMALL menus. We analyzed the ranks assigned to menus to reflect ease of finding and selecting items using Friedman ANOVA (nonparametric equivalent of RM ANOVA). One participant completed the questionnaire incorrectly so we excluded it from the analysis. Although participants did not express that finding items was easier with any of the menus (χ^2=5.34, df=3, p=.15, N=17), they subjectively perceived the menus different with respect to easiness of selecting (χ^2=12.44, df=3, p=.006, N=17). Sign Test comparisons between the menus revealed that each radial menu was perceived as faster than the CS menu (CRL-CS p=.027, SPARSE-CS p=.04, FAN-CS p=.027).

For menu preference orders, we analyzed the frequencies with which each menu was chosen first. The Chi-square statistic did not reach significance (p=.27). The most popular choice was FAN.

BIG menus. For each of the four self-reported variables (see section 4.5) we analyzed the frequency with which each condition (CS better, CRL better, no difference) was chosen first by calculating the Chi-square statistic to determine if actual frequencies were significantly different from the case in which all frequencies are equal. Interestingly, the Chi-square statistic did not reach significance for questions related to speed and ease of finding items. However, the participants expressed a strong preference for ease (χ^2(2)=25.3, p<.001) and speed (χ^2(2)=13, p=.002) of selections, CRL being the most popular choice. There were no significant correlations between any of the variables.

4.8.5 Errors

Errors were analyzed together for both parts of the experiment. For each participant we aggregated the number of errors (i.e. clicks missing the target) made with each menu. The result of Friedman ANOVA was not significant. A total of 3.6% of all the trials were erroneous.

4.9 Screen usage

To estimate the screen usage for the radial menus, we calculated a bounding box (and a bounding circle) size that totally enclosed the menu for each level. These two measures are suitable for radial layout if the levels are fully populated. To measure space consumption for smaller menus, we summed the areas of all visible items. For the cascading menus, we summed the sizes of the submenu boxes for each level (see Table 1). Screen

Table 1 Screen usage measured in pixels.

	level 1	level 2	level 3
CS-BIG$_{PIXELS}$	28.800	77.760	146.880
CRL-BIG$_{PIXELS}$	12.168	35.724	74.654
CRL-BIG$_{BOX}$	22.499	77.006	161.202
CRL-BIG$_{CIRCLE}$	19.113	62.458	129.462
CS-SMALL$_{PIXELS}$	24.768	49.536	74.304
CRL-SMALL$_{PIXELS}$	10.647	22.425	36.165

consumption for the radial menus is lower than for the CS menu by roughly 50% in terms of screen pixels consumed.

5. DISCUSSION

5.1 Visual search and pointing

Visual search time. The experimental results show that users searching for commands find them faster in a CS menu than in any radial menu. The advantage is larger for small menus (31%) and smaller for bigger menus (14%). For small menus the CS's 31% advantage is roughly the same on all levels. For big menus the advantage decreases to 10% (138ms) on the first two levels but reaches 20% (425ms) on the third level. The CS advantage for visual search is likely due to shorter visual scan paths and parallel visual processing [9]. Using different techniques for laying out the menu items around the rings (SPARSE, FAN, CRL) does not have any significant effect on search time. However, among the radial menus the CRL menu provides the lowest (and SPARSE the biggest) time difference between first and the third levels.

Pointing time. The measurements for the pointing time are reversed: the users perform faster with radial menus than with CS menus with no exceptions – small and big menu sizes and on all three levels. The advantage is similar for two menu sizes (34%). Again, the CRL menu provides the lowest time difference between first and third levels – this time compared to all other menus. This is emphasized by the CRL menu being the fastest menu on the third level. This suggests also that the CRL design shortens the distances between the levels most effectively.

5.2 Radial menus (RM) vs CS

RM vs CS for novices. The task used in the experiment simulated novices (i.e. participants always had to search for the items on each level and any learning effects were prevented). Therefore, we can compare novice performance of CS and RM using total selection times. It turns out that the overall selection time for CS and CRL menus do not differ from each other. The CS menu, however, is 10% faster than FAN and SPARSE menus. CRL compensates slower visual search with faster selections. So for novice users, the CS and CRL menus provide similar performance. However, if the selection tasks become more visually demanding than the one used in the current experiment, it is likely that the visual search advantage of the CS menus will dominate the overall selection time.

RM vs CS for experts. In contrast to novices, the menu selection performance for experts is a combination of pointing and decision times – search time can be disregarded since experts know the position of menu items. This is based on extensive empirical evidence [7, 16] and allows for the following calculations. According to [7, 8, 11, 16], the decision time is a logarithmic function of the number of items the user has to choose from. Therefore, decision time for the CS and the RM is similar and can be disregarded for a qualitative assessment of CS and radial menus for experts. However, for a quantitative assessment, a decision time offset has to be added to the pointing time. For experts, a decision has to be made before selection on each level [17]. According to [7], the decision times for 6, 7, 12, and 17 items are 447ms, 465ms, 527ms and 567ms respectively. When adding the decision time offsets to the pointing times it turns out that the overall selection time is faster by approximately 17% for the RM regardless of the menu level and size (small or big). Our result supports prior evidence where a single level radial menu was found to be 15% faster than linear menu for experts [5].

RM vs CS for dedicated users. Experts have to decide on each level which item to select. However, dedicated users [6], who use menus more extensively than experts, need to decide only once at the beginning of the selection which item to select [17]. In this case, they have to choose from all items in the menu. According to [7], the decision time for 1428 items (our big menu) is 1078ms, and for 216 items (small menu) 860ms. Adding these offsets to the pointing times result in RM being faster by approximately 20% regardless of the menu level and size (small or big).

If the selection tasks become more demanding in terms of pointing than the one used in the current experiment, it is likely that the pointing time advantage of the radial menus will dominate the overall selection time even more. Furthermore, given that the CS menus used in the experiments provide only a lower bound for the required pointing time, the advantage of radial menus are likely to increase if CS menu variants that require precise steering or dwelling are used (see section 4.2).

Using Point-completion deadline technique (PCD). The PCD technique is an accepted, legitimate technique used to compare layouts in selection tasks [10]. However, there is a concern that this technique forces participants into unnatural behavior since they are forbidden to move the mouse while searching.

Experts and dedicated users. Parallel searching and pointing is not of concern for experts and dedicated users since they do not search but point straight to a remembered item location [7, 12, 16]

Novices. In [7] participants were able to move the mouse while searching, but the proposed predictive model assumed that pointing starts after finding an item (as in our study). *"The predictions were accurate within 2% of empirical data"* [7] (4 designs tested). A similar result was found in [4] with the ACT-R/PM model.

Consequently, moving the mouse while searching does not have any practical effect when comparing menu designs.

5.3 Subjective measurements

The participants' perception is in agreement with the objective results for novice performance (i.e. no preference for any of the menus) and pointing times (i.e. the radial menus are faster). However, although finding items is faster with the CS menu the participants' perception is that CS and radial menus do not differ.

Participants favoring the radial menus appreciated blank spaces separating the items: "*Although you can accommodate less items, it seems easier to find/locate items in radial version and the 'white space' gives an uncluttered feeling which leads to a perception of it being easier to use" or "It is not convenient going through long lists of items (cascading menu) and the letters mix in the eyes. In the radial menu I was faster and letters are not so mixed since there is blank space between the items*".

Participants in favor of the CS menu stress its predictability and familiarity in contrast to the 'strange' radial layout. This will probably change as users get more familiar with radial menus.

The analysis of the interviews revealed two design considerations we were not aware of. One participant says (big menu size): "*In radial – the first two levels are very good – better than cascading. The last one is worse than cascading – my eyes needed to do too big moves*". Some people complain also about distraction caused by the adjacent items from inner levels: "*the previous levels are 'in the way' geographically of the outer level, causing more confusion when searching for the target as opposed to the cascading layout that produces a visual separation on areas of search (the rectangular areas that contain only the current level)*". This suggests that dimming intensity of the inner rings could be a possible improvement for large radial menus.

6. CONCLUSIONS

In this paper, we presented a design of a hierarchical radial menu (CRL menu) and a study investigating how radial layout influences two main factors governing menu selections – searching for menu items and pointing to menu items.

Visual search is faster with linear layout (CS menu). The advantage is larger for small menus (31%) and smaller for bigger menus (14%). Pointing is faster with radial layout (CRL menu). The advantage is similar for small and big menu size and on all three levels (34%).

The results demonstrate that radial layout is beneficial for the design of displayed hierarchical menus.

The best performing radial menu (CRL menu) did not differ from the CS menu with respect to novice performance. However, if the task becomes more visually demanding than the one used in the current experiment, the CS menu is likely to gain an advantage. This suggests, for example, that using radial layout for menus with changing content may worsen their performance. However, despite slower visual search for the CRL menu, many participants expressed it was easier to search through its 'not so mixed' items. Searching the third level ring full of items (big menu) becomes particularly slow for the CRL menu. One potential improvement, in this context, might be to dim the inner rings in order to decrease their visual interference with the outermost ring.

The CRL menu provided the best expert and dedicated user performance for both menu sizes. The CRL menu will be particularly beneficial for professional applications where users invoke hundreds of commands per session (this is also the case for computer games). Some applications may consider using the CRL menu as a quick way of accessing a customized set of commands. If the selection tasks become more demanding in terms of pointing than the one used in the current experiment, it is likely that the advantage of the radial menus will increase. This suggests, for example, that the CRL menu will gain an additional advantage in menus where multiple selections are possible.

The CRL menu uses less screen space than the CS menu. We recommend using the CRL menu as a context menu to avoid repositioning and slow round-trips to a menu bar. It will be most effective with short command labels (from one to three words).

In this paper, we demonstrated also that the point-completion deadline is a useful technique for evaluating menu designs.

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